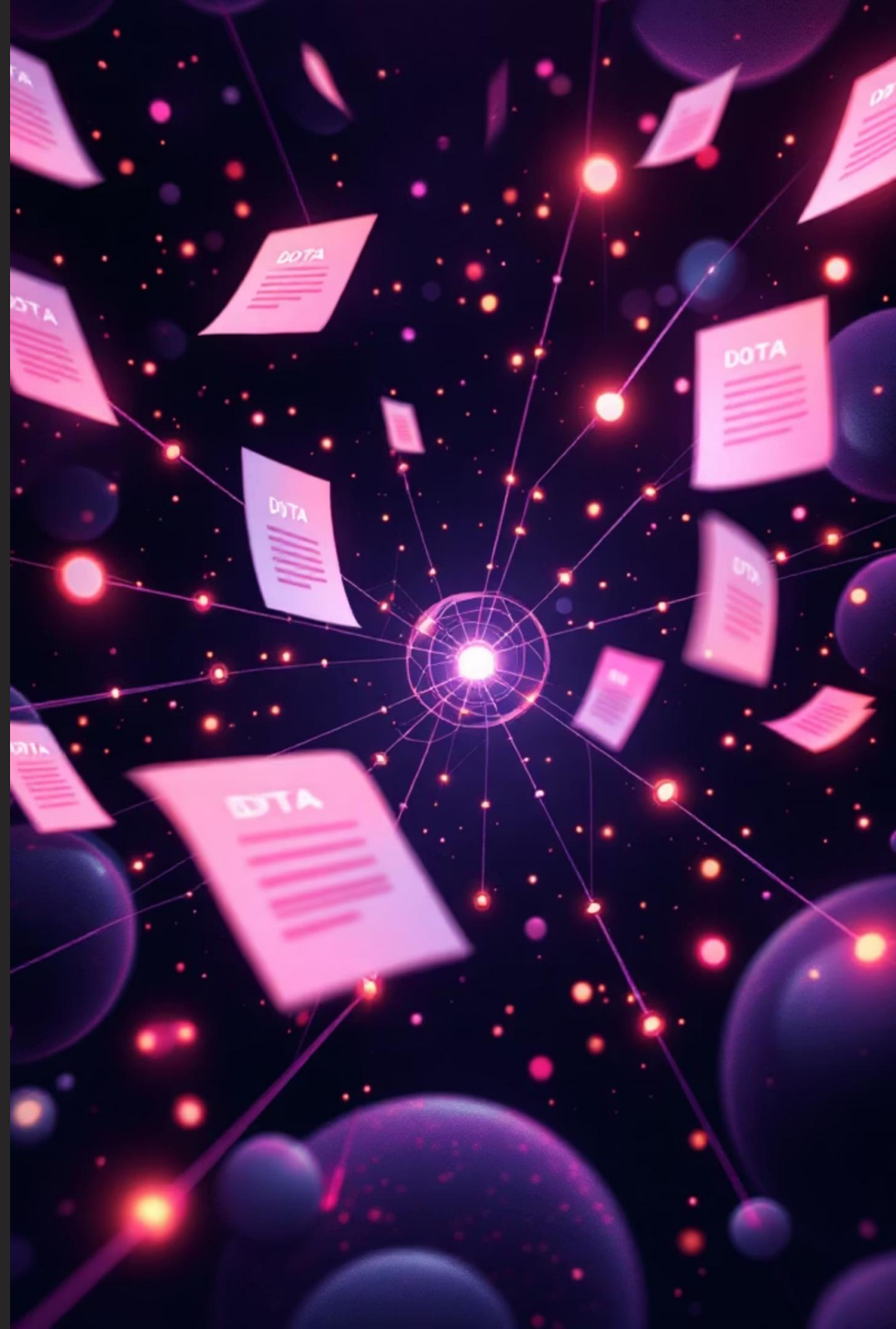
Document Similarity Checker using Machine Learning

A smart, fast solution for comparing text documents using TF-IDF and Cosine Similarity



Problem & Objective

The Problem

Manual Comparison is Hard

Comparing large documents by hand is error-prone and inefficient.

Time-Consuming

Reviewing hundreds of lines of text manually is not scalable.

The Objective

Measure Similarity

Automatically compare two text documents and quantify their resemblance.

Generate a Score

Produce a clear similarity percentage using machine learning techniques.

Input — What the System Takes

Two Text Documents

User provides both documents directly — no dataset or training data required.

Real-Time Processing

Results are generated instantly upon input, making the system fast and interactive.

Example

Doc 1: Machine learning is fun
Doc 2: Machine learning is very interesting

Model & Working

The pipeline converts raw text into meaningful numerical representations, then measures the angular distance between them to produce a score.

Techniques Used

TF-IDF

Weights words by their importance across documents, not just frequency.

Cosine Similarity

Measures the angle between two vectors — closer to 1 means more similar.

- Based on **word importance**, not exact string matching — making it robust to paraphrasing.

Result — What the System Outputs

78%

Similarity Score

Example output for the two sample documents provided

0% – 40% Low Similarity	**41% – 70%** Moderately Similar
71% – 100% Highly Similar	

Advantages & Limitations

✅ Advantages

Simple & Fast

Lightweight implementation with near-instant results.

Works on Any Text

Domain-agnostic — handles legal, academic, or casual documents.

Easy to Implement

Built on well-established, minimal Python libraries.

⚠ Limitations

No Deep Understanding

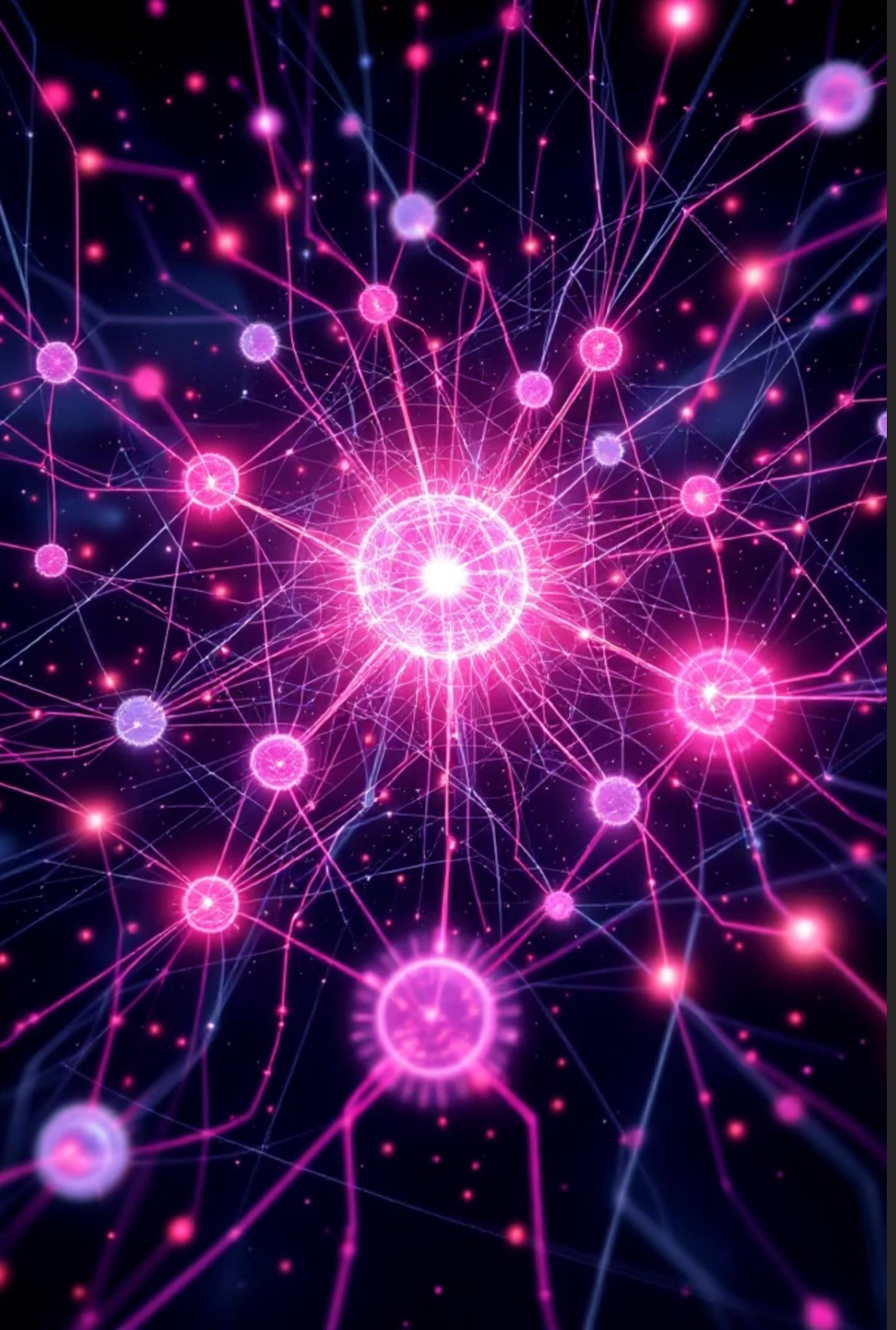
Cannot grasp context, tone, or intent behind words.

Word-Level Only

Synonyms and paraphrasing can reduce accuracy.

Struggles with Complexity

Less reliable for nuanced or highly technical sentences.



Conclusion

This project demonstrates how classic machine learning techniques can solve a real-world problem efficiently and elegantly.

01

Effective Document Comparison

Uses TF-IDF and Cosine Similarity to reliably measure text resemblance.

02

Fast & Accessible

A lightweight solution that runs in real-time without large datasets.

03

Future Scope

Can be enhanced using advanced NLP models like BERT or Word2Vec for deeper semantic understanding.